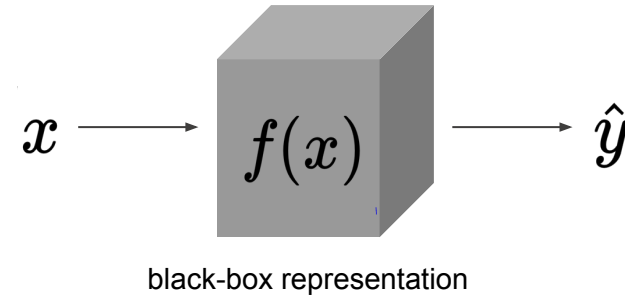


A review of some important concepts in deep neural networks

Deep Neural Networks

$$f = H_1(H_2(H_3 \dots H_n(x)))$$

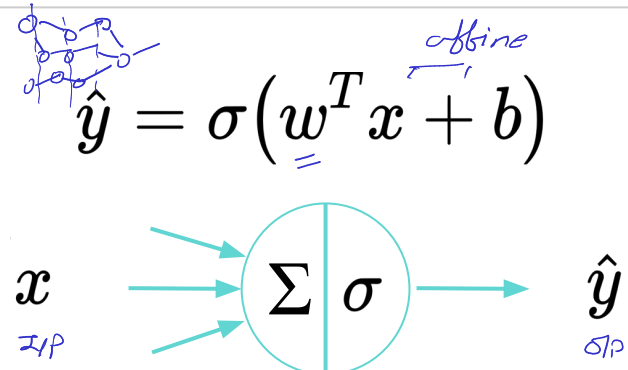
- compute a **composition of functions** $\Rightarrow RNNs$
- linear transformations** and **nonlinearity** $\Rightarrow RNNs$
- can approximate real continuous functions
 - ✓ theoretical: universal approximation theorem
 - ✓ practical: **depends on data distribution, training**
- deep networks are considered as a black-box



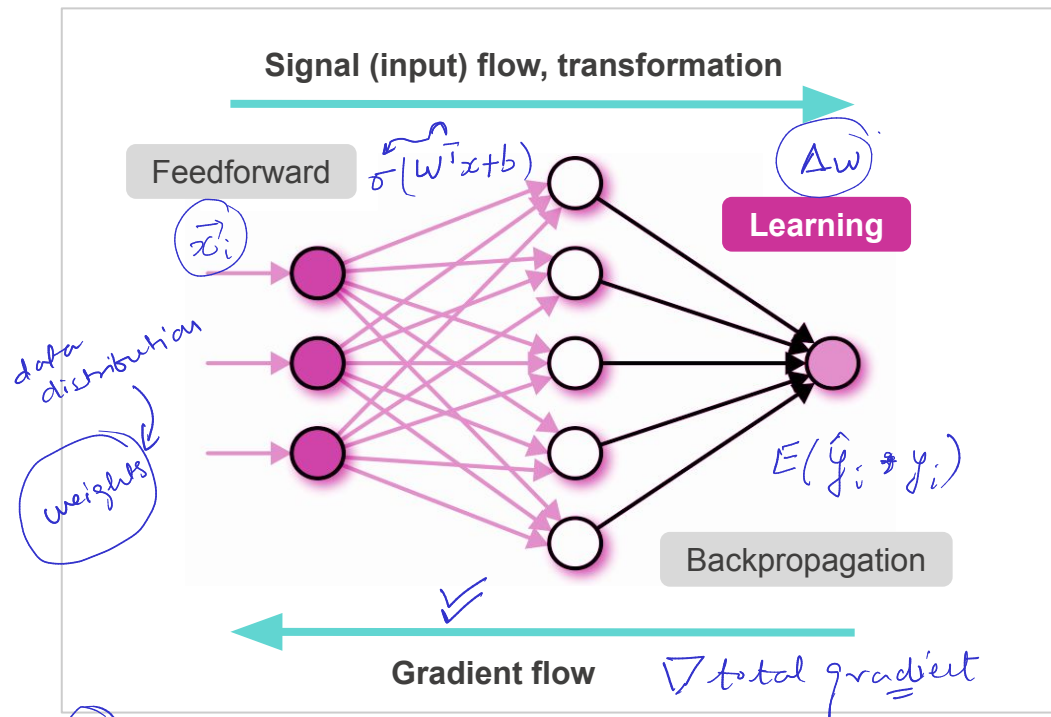
For example, in a **supervised learning** problem:

- networks model the **map** from inputs to outputs
- model is a reasonable **approximation** of true map
- network learns **representations** across layers
 - from low-level features to high-level abstractions
 - CNN (edges, contours, objects, ... concepts)

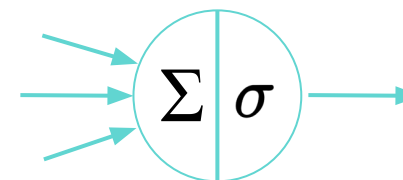
pixels $\uparrow$ input $\Rightarrow$ raw data



A review of some important concepts in deep neural networks



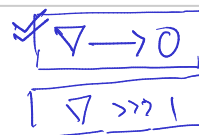
- represent **computation graphs**
 - nodes are computation elements
 - edges are information flow channels
 - computation nodes are differentiable



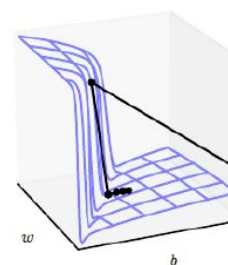
- representation of information
 - weights are **distributed** across the network
 - **hardware-efficient trainability** is desired

Vanishing and Exploding Gradients

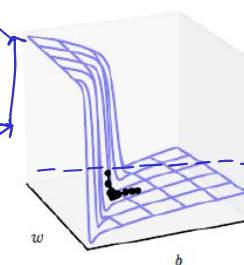
- Gradient instability is a problem for network trainability
 - Vanishing gradients tend to zero, layers learn "slowly"
 - Exploding gradients offshoot, layers fail to converge



No clipping



clipping



threshold $\nabla_f < \nabla$
 $\Downarrow$
 ∇_f

stable